

Literature Survey

Blockchain Technology for Land Registration Applications

1. Scope

This survey covers 15 papers on blockchain-based land registration and administration, spanning systematic reviews, country-specific implementations, and technical architecture proposals published between 2022 and 2026. It is organized to support a project's related-work / literature review chapter.

2. Literature Table

No.	Title	APA Citation	Link
1	Blockchain Technology in Lands Registration: A Systematic Literature Review	Zein, R. M., & Twinomurinzi, H. (2023). JeDEM – eJournal of eDemocracy and Open Government, 15(1), 1–36.	https://doi.org/10.29379/jedem.v15i1...
2	Benefits and Challenges of Blockchain Technology in Real Estate: A Systematic Literature Review	(2026). Real Estate, 3(2), Article 6. Reviews peer-reviewed publications 2016-2025.	https://doi.org/10.3390/realestate30...
3	Blockchain in Real Estate: Recent Developments and Empirical Applications	(2022). Land Use Policy. Systematic review of 262 documents across land administration, transactions, tokenization, and management.	https://www.sciencedirect.com/scienc...
4	A Systematic Literature Review on Blockchain for Real Estate Transactions: Benefits, Challenges, Enablers, and Inhibitors	(2023). Identifies lack of regulation, lack of knowledge, and adoption resistance as leading inhibitors.	https://www.researchgate.net/publica...
5	Practicality of Blockchain Technology for Land Registration: A Namibian Case Study	(2025). Land, 14(8), 1626.	https://doi.org/10.3390/land14081626
6	Land Registration and Inheritance Automation System Using Blockchain (LRIAS)	(2025). Engineering Proceedings, 107(1), 97. Focus: Pakistan, women's inheritance rights.	https://doi.org/10.3390/engproc20251...
7	Securing Land Transactions using a Blockchain and GIS Enabled Biometric Land Information System	Tonoy, B. B., Roy, G., Hasan, M., & Hasan, M. (2026). Conference Paper.	https://www.researchgate.net/publica...
8	Blockchain-Based Framework for Secure and Reliable Land Registry System	Khan, R. H., & Salahuddin, S. S. (2025).	https://www.researchgate.net/publica...
9	LandChain: A Blockchain Based Secured Land Registration System	(2026). IEEE Conference Publication. Focus: Bangladesh.	https://ieeexplore.ieee.org/document...
10	Using Blockchain to Overcome the Issues in Land Registry Management: A Systematic Review	(2025). IEEE Conference Publication.	https://ieeexplore.ieee.org/document...
11	BACP-LRS: Blockchain and IPFS-Based Land Record System	(2025). Focus: India; addresses inconsistent data and lack of transparency using Ethereum + IPFS.	https://www.researchgate.net/publica...

No.	Title	APA Citation	Link
12	Cadastral: Blockchain-Based Land Registration System	Suresh, S., Van, K., Yadav, C., Jacob, M., & Wangikar, V. (2025). In Blockchain – ICBC 2024, LNCS vol. 15425. Springer, Cham.	https://doi.org/10.1007/978-3-031-77...
13	Secured Land Registration Using Ethereum Blockchain and IPFS	(2025). IEEE Conference Publication. Focus: India; uses Solidity, IPFS, Sepolia TestNet.	https://ieeexplore.ieee.org/document...
14	Smart Land Registry System Using Blockchain: Challenges and Solutions	(2025). Springer book chapter. Compares public, private, hybrid, and consortium blockchain configurations.	https://doi.org/10.1007/978-3-031-87...
15	Smart Contract-Based Land Registration System Using Blockchain	(2025). IEEE Conference Publication.	https://ieeexplore.ieee.org/document...

3. Thematic Synthesis

Grouping the papers by theme (rather than reading them one at a time) surfaces the field's shared assumptions and where individual papers diverge from them:

Theme	Representative Papers	Core Finding
Systematic reviews / meta-syntheses	Papers 1-4	Consistent themes across the field: transparency, fraud reduction, and disintermediation are the dominant motivations; lack of regulation, legacy-system integration, and low digital literacy are the dominant barriers.
Developing-country case studies	Papers 5, 6, 9, 11, 13	Namibia, Pakistan, Bangladesh, and India studies all report the same underlying problem — fragmented paper-based registries prone to fraud — and converge on Ethereum + smart contracts as the default technical solution.
Technical architecture (IPFS, GIS, biometrics)	Papers 7, 11, 12	Later papers extend basic Ethereum + smart-contract designs with off-chain storage (IPFS) for documents/images and biometric or GIS layers for identity and spatial verification — addressing the on-chain storage-cost and identity-fraud limitations of earlier designs.
Governance / configuration choices	Paper 14	Highlights that public, private, hybrid, and consortium blockchain configurations trade off differently between transparency, throughput, and government control — a design decision most implementation papers skip over.
Social-equity angle	Paper 6	The Pakistan LRIAS system is the only paper in this set framing blockchain land registration explicitly around gender-equitable inheritance rights, rather than efficiency/fraud alone.

4. Common Problem Statement Across the Literature

Nearly every paper in this set opens with the same diagnosis of the conventional land registry: paper-based or siloed government databases that are prone to forgery, slow to update, opaque to buyers, and dependent on intermediaries (notaries, registrars, brokers) whose involvement adds cost, delay, and corruption risk. Blockchain is proposed as a shared, tamper-evident, and disintermediated alternative — a near-universal framing across Papers 1-15.

5. Recurring Technical Architecture

1. Ethereum as the dominant platform — used in Papers 5, 7, 9, 11, 12, 13, chosen for its mature smart-contract tooling (Solidity, Truffle/Ganache, MetaMask) even though it is not purpose-built for government-scale record systems.
2. Smart contracts for transaction logic — automate ownership transfer, buyer/seller verification, and land-inspector approval workflows (Papers 5, 9, 12, 15), reducing manual intermediation.
3. Off-chain storage via IPFS — because storing full documents/images on-chain is costly, several papers (7, 11) offload deeds and certificates to IPFS and store only hashes on-chain.
4. Identity and spatial verification layers — GIS mapping and biometric authentication (Paper 7) are added on top of the base blockchain layer to solve the 'who owns which exact plot' problem that pure ledger transparency does not solve by itself.

6. Gaps Identified Across the Literature

- Legal-technical disconnect — Papers 1 and 4 both flag that smart-contract automation is being designed largely independent of each country's actual property law and registration statutes, creating solutions that may not be legally enforceable as built.
- Scale and pilot-stage limitation — Paper 3's review of 262 documents found that most real-world applications remain small-scale pilots or add-on layers to existing systems, not full production replacements.
- Regulatory and institutional barriers dominate over technical ones — Paper 4 ranks 'lack of regulation' as the leading inhibitor, ahead of any purely technical limitation, suggesting future work should weigh governance design as heavily as system architecture.
- Equity dimension is under-explored — only Paper 6 frames blockchain land registration around a social-equity outcome (women's inheritance rights); most other papers optimize for speed/fraud-reduction only.
- Interoperability with legacy government databases — repeatedly named as a practical blocker (Paper 1) but rarely addressed as a first-class design problem in the technical papers.

7. Suggested Use in Your Report

- Open your related-work section with Papers 1-4 (the systematic reviews) to establish the field's overall shape before narrowing into specific implementations.
- Use Papers 5, 6, 9, 11, 13 as country-context precedents if your project targets a specific region.
- Use Papers 7, 12 to justify any architecture choices beyond a bare Ethereum + smart-contract setup (e.g., adding IPFS or biometric/GIS layers).
- Use the gap list in Section 6 directly as your project's motivation / problem-statement section — each gap is a legitimate, citable opening for a novel contribution.